## DIAGNOSIS AND TESTING

### A / C PRESSURE TRANSDUCER

The A/C pressure transducer is tested using a DRBIII® scan tool. Refer to the appropriate diagnostic information. Before testing the A/C pressure transducer, be certain that the transducer wire harness connection is clean of corrosion and properly connected. For the A/C to operate, an A/C pressure transducer voltage reading between 0.451 and 4.519 volts is required. Voltages outside this range indicate a low or high refrigerant system pressure condition to the powertrain control module (PCM). The PCM is programmed to respond to a low or high refrigerant system pressure by suppressing operation of the A/C compressor. Refer to the A/C Pressure Transducer Voltage chart for the possible conditions indicated by the transducer voltage reading.

*A/C PRESSURE TRANSDUCER VOLTAGE*

| Voltage | Possible Indication |
|---|---|
| 0.0 | 1. No sensor supply voltage from PCM.<br>2. Shorted sensor circuit.<br>3. Faulty transducer. |
| 0.150 TO 0.450 | 1. Ambient temperature below 10° C (50° F).<br>2. Low refrigerant system pressure. |
| 0.451 TO 4.519 | 1. Normal refrigerant system pressure. |
| 4.520 TO 4.850 | 1. High refrigerant system pressure. |
| 5.0 | 1. Open sensor circuit.<br>2. Faulty transducer. |

### REMOVAL

**NOTE: Note: It is not necessary to discharge the refrigerant system to replace the A/C pressure transducer.**

(1) Disconnect and isolate the negative battery cable.

(2) Disconnect the wire harness connector from the A/C pressure transducer (Fig. 21).

(3) Remove the A/C pressure transducer from the A/C discharge line and remove and discard the O-ring seal.

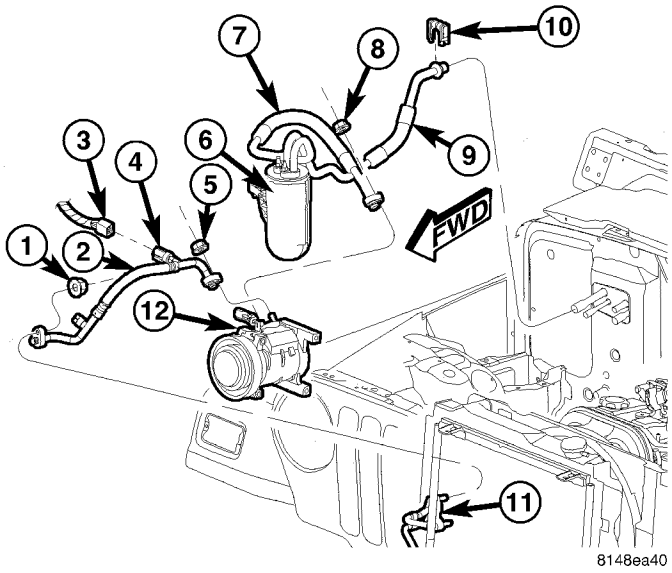


***Fig. 21 A/C Pressure Transducer - 2.4L Engine***

1 - NUT
2 - A/C DISCHARGE LINE
3 - WIRE HARNESS CONNECTOR
4 - A/C PRESSURE TRANSDUCER
5 - NUT
6 - ACCUMULATOR
7 - A/C SUCTION LINE (FRONT SECTION)
8 - NUT (4)
9 - A/C SUCTION LINE (REAR SECTION)
10 - SECONDARY RETAINING CLIP
11 - A/C CONDENSER
12 - A/C COMPRESSOR

### INSTALLATION

**NOTE: Use only the specified O-ring as it is made of special material for R-134a. Use only refrigerant oil of the type required for the A/C compressor.**

(1) Lubricate a new rubber O-ring seal with clean refrigerant oil and install it on the A/C pressure transducer fitting.

(2) Install the A/C pressure transducer onto the A/C discharge line. Hand-tighten the transducer securely.

(3) Connect the wire harness connector to the A/C pressure transducer.

(4) Reconnect the negative battery cable.

## BLEND DOOR ACTUATOR

### DESCRIPTION

The blend door actuator is a reversible, 12-volt direct current (DC), servo motor which mechanically positions the blend-air door (Fig. 22). The blend door actuator is located on the bottom of the HVAC housing.

The blend door actuator is contained within a black molded plastic housing with an integral wire connector receptacle. The actuator also has an output